

GAURAV BOOB

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SUMMARY

A motivated professional with experience in two reputed software companies, aspiring to pursue advanced studies abroad and leverage my technical skills and industry knowledge.

EDUCATION

B.E. - Electronics & Telecommunication Aug 2020 - Aug 2024
SCTR's Pune Institute of Computer Technology, GPA : 8.94

SKILLS

- Python (NumPy, Pandas, Scikit-learn)
- Java, C++, Spring Boot
- Power BI and Tableau
- GIT, SQL
- TensorFlow, Deep Learning, Computer Vision, Open CV
- Katalon
- Critical Thinking

EXPERIENCE

Fidelity Information Services (FIS), Pune Jan 25 - Ongoing

Software Engineer

- Contributing to the Payments team, focusing on US payment methods like Wire transfers, and ACH.
- Led enhancements for State Street Bank by integrating new recipient SWIFT code fields across the payment import pipeline, including backend services, database integration, and Angular-based UI components.
- Collaborating with DBAs to optimize data retrieval queries for improved performance and scalability in high-volume payment workflows.
- Enhanced international wire payments by adding support for additional fields (e.g., Sender-to-Receiver information) during payment import and template loader processing.
- Implemented robust validations and business logic to ensure data integrity and compliance in payment processing.

Dragonfly Financial Technologies (Acquired by FIS) July 24 - Dec 24

Associate Software Engineer

- Implemented sensitive data logging fixes, including account number and details masking, across 50+ key modules to meet compliance and security standards.
- Enhanced backend solutions and optimized APIs, resulting in a 30% improvement in response time for transactions across the payments platform while ensuring system reliability during peak usage periods.

Dragonfly Financial Technologies Jan 24 - March 24

Associate Software Engineer Intern

- Worked closely with DBAs to identify performance bottlenecks, implementing query optimizations that reduced execution time by 27% and hence leading to faster data retrieval and improved overall system responsiveness.
- Wrote Katalon automation scripts for faster, more accurate releases. Enhanced an in-house automation tool, improving test coverage and reducing manual effort.

SCTR's Pune Institute of Computer Technology Jan 23 - May 24

Undergraduate Research Assistant

- An extensive research experience in the Deep Learning lab at PICT, focusing on the classification of Diabetic Retinopathy. Implemented pre-processing techniques such as CLAHE, Subtract Mean Processing, etc. Fine tuned YOLOv8x model, achieving a classification accuracy of 95%. Research work is currently under consideration for publication.

Vasundharaa Geo Technologies Pvt. Ltd. Sep 22- Nov 22

Data Science Intern

- Led advanced tech stack utilization, employing Computer Vision and Image Segmentation techniques such as U-Nets to achieve 87% mAP for semantic mask generation.
- Conducted extensive research on image pre-processing techniques, including normalization and augmentation, resulting in a 25% reduction in processing time for satellite images, which improved workflow efficiency and data analysis turnaround.

PUBLICATIONS

- Leveraging Deep Learning for Lip Reading: A Comprehensive Analysis, Springer Conference (https://link.springer.com/chapter/10.1007/978-981-97-8591-9_30)
- Lipreading for Indian Accent English is in peer review process, Springer Journal (In Progress)
- Classification of Diabetic Retinopathy, Springer Journal (In Progress)

PROJECTS

Lipreading for Indian Accent English | [Demo](#)

(Research Project) May 2024

- Trained a custom CNN-BiLSTM model to interpret speech from visual cues, focusing on Indian Accent English. Achieved a CER of 4% and a WER of 15%, ranked first position in an international level project competition. Developed a novel Indian English accent dataset consisting of 500 videos along with their alignment files. Research work is currently under consideration for publication at Signal, Image and Video Processing Springer journal.

Chat with Scientific Documents | [GitHub](#)

March 2024

- Fine-tuned a Visual Transformer model namely Nougat for parsing scientific documents in any electric form. The model was specifically trained to handle complex document structures, including mathematical equations, tables, and figures, ensuring minimal loss of semantic information during Optical Character Recognition (OCR) tasks. Integrated with Gemini 1.0 Pro for intuitive user interaction and query answering.

Jigsaw Puzzle Solver | [GitHub](#)

Jan 2023

- Implemented a Jigsaw puzzle solver using custom CNNs. Trained the solver for accurate 6x6 puzzle piece permutation prediction and placement. Enhanced spatial awareness for accurate edge detection and placement, improving overall prediction performance. Applied permutation learning concept.

Classification of Diabetic Retinopathy

(Research Project) Jan 2023-May 2024

- I developed a computer model leveraging CNN and LSTM to evaluate how severe diabetic eye disease is based on retina images. By employing data cleaning, refining images, and implementing balancing methods, I was able to achieve over 95% accuracy on sizable datasets. We played a key role in identifying both the early and advanced stages of the disease, along with detecting swelling in the retina, known as macular edema. We tested our models in real hospital settings, and they outperformed previous methods. Additionally, we looked into predicting risks and detailing features of eye damage to help doctors better manage their patients.

CERTIFICATIONS

- Full Stack Development : Java + JSP + Spring + Boot + JS + React - Study Easy Organization, Chand Sheikh, Udemy
- Python for Data Science, AI & Development, Coursera
- Share Data Through the Art of Visualization, Coursera
- Applied Data Science with Python Specialization, UMich, Coursera, Ongoing
- IBM Data Science Professional Certificate, Coursera, Ongoing
- Google Data Analytics Professional Certificate, Coursera, Ongoing

COCURRICULARS

1. Frontend Web Developer at IEEE Pune

May 2022 - May 2023

- Volunteered to develop & enhance the front-end of the IEEE Pune Section website, actively contributing to impactful projects like AISYWLC and other IEEE initiatives to engage and support the tech community.
- Improved website performance, achieving a 40% acceleration in page load times and boosting user engagement through a responsive, dynamic interface built with ReactJS.

2. Podcast Technical Lead IEEE Pune Section

May 2021 - Mar 2022

- Volunteered as the Technical Lead for the podcast team, where I collaborated with a multidisciplinary team to produce engaging episodes and manage technical setups.
- This role provided a platform to network with prominent guests from diverse domains, enriching my understanding of various fields and honing both my technical and interpersonal skills.

Hackathons:

1. Winner at INC 2024 Hackathon

April 2024

Issued by SCTR Pune Institute of Computer Technology · Apr 2024 Our team earned the winning position in the Digital Speech Processing category at INC'24 (Impetus and Concepts) the annual flagship event of PICT, competing against 200+ teams.

2. Winner at MINEd Hackathon'24

Feb 2024

Issued by Binghamton University & Nirma University · Feb 2024 Our project aimed to develop a robust framework to parse multiple formats beyond PDF, integrated with language models for intuitive user interaction and query answering which will streamline document analysis and knowledge dissemination, fostering a more efficient ecosystem for research and collaboration. We were the winners among 150 participating teams.